

(Q)SAR Model Reporting Format (QMRF) v2.1

Human estrogen receptor alpha (ER α) direct-binding classification ER Predictor model v7

Model ID: erba_classification_v7_er_alpha

QMRF drafting date: 20 August 2026

Document status: Draft 1.0

CRITICAL STATUS: This QMRF documents the current frozen model accurately. The current endpoint combines heterogeneous public direct-binding assay evidence. The 224-compound sensitivity partition is historically exposed internal evidence, not an external validation set. This model and its performance must not be described as externally or regulatorily validated. K-FDA's future single-assay dataset should remain sealed for prospective external validation unless K-FDA explicitly authorizes another use.

Administrative completion checklist

Item	Status before K-FDA submission
QMRF author name, institution, address, email	OWNER INPUT REQUIRED
Model developer names and contact details	OWNER INPUT REQUIRED
Public availability/redaction decision for training records	OWNER DECISION REQUIRED
Exact assay-method contract from K-FDA	PENDING K-FDA DATA
Prospective external validation results	NOT YET AVAILABLE
Scientific-owner sign-off and document version	OWNER APPROVAL REQUIRED

1. QSAR identifier

1.1 QSAR identifier (title)

Human estrogen receptor alpha (ER α) direct-binding classification using an RDKit path fingerprint and radial-basis-function support vector classifier — ER Predictor model v7 (erba_classification_v7_er_alpha).

1.2 Other related models

The ERTA activity model distributed in the same desktop application is a distinct endpoint/model and is not part of this QMRF. Historical ER β and ER α regression research artifacts are outside the current FDA-facing product scope. VEGA Estrogen Receptor Relative Binding Affinity and VEGA/CERAPP Estrogen Receptor-mediated effect models were consulted only as examples of QMRF reporting; they are not components of this model.

1.3 Software coding the model

ER Predictor, standalone Windows desktop application. The frozen classifier is serialized as production_classification_model_ERalpha.joblib and executed through a scikit-learn Pipeline. The distributed application bundles Python and required libraries; no network connection is needed for direct-SMILES prediction. Optional PubChem lookup is used only to resolve CAS identifiers to structures and is separate from the model algorithm.

2. General information

2.0 Abstract

This binary QSAR estimates whether an organic chemical belongs to the ER α direct-binding positive or negative class. It was developed from curated public records dominated by ChEMBL, with a smaller PubChem BioAssay contribution. The current development set pools multiple direct-binding assay endpoints and protocols; it is therefore an assay-

heterogeneous binding classifier, not a model of one standardized assay method. SMILES are parsed and sanitized, reduced to the fragment selected by RDKit LargestFragmentChooser(), rejected unless that retained fragment contains carbon and only supported elements, and converted to canonical isomeric SMILES. The retained structure is represented by a 2,048-bit RDKit topological fingerprint and classified by an RBF support vector machine. Candidate selection used nested scaffold/identity-connected grouped cross-validation optimized for Matthews correlation coefficient (MCC) at a fixed probability threshold of 0.5. Applicability domain (AD) is an operational reliability aid combining standardized 5-nearest-neighbour distance and fingerprint similarity to the 1,481-compound development reference. Validation evidence is internal and historically exposed; no independent external validation is currently available.

2.1 Date of QMRF

20 August 2026.

2.2 QMRF author(s) and contact details

[TO BE COMPLETED BY SCIENTIFIC OWNER: full name, institution, postal address, email, ORCID if applicable].

2.3 Date of QMRF update(s)

Not applicable — first draft.

2.4 QMRF update(s)

Not applicable — first draft. A retrained model, changed training set, changed assay definition, changed descriptor, changed algorithm, or changed decision threshold requires a new model identifier and a new QMRF rather than an administrative update to this document.

2.5 Model developer(s) and contact details

[TO BE COMPLETED BY SCIENTIFIC OWNER: developer names and contact details]. Computational implementation repository: private repository phdgil/er-predictor-kfda; release access controlled by the owner.

2.6 Date of model development and/or publication

2026. The frozen model bytes and evaluator-bound release manifest identify model version v7.

2.7 References to main scientific papers and/or software package

- No peer-reviewed scientific paper specifically describing this exact model version is currently available.
- RDKit: open-source cheminformatics toolkit, version 2024.03.5 at model development/runtime.
- scikit-learn: machine-learning library, version 1.5.2 at model development/runtime.
- Internal model evidence: protocol.json, production_classification_model_ERalpha_manifest.json, release_manifest_ERalpha.json, nested_cv_results.csv, nested_cv_outer_predictions.csv, and internal_resplit_sensitivity_evaluation.json in classification_modeling_nested_mcc_v7.

2.8 Availability of information about the model

The algorithm, preprocessing policy, feature generator, model artifact hash, split protocol, and aggregate/prediction-level internal validation evidence are available to the project owner. Source database records are public-source-derived, but redistribution of row-level records and identifiers must follow the originating databases' terms and the project's disclosure decision. The packaged classifier is distributed with the private/internal ER Predictor release. This QMRF does not itself authorize unrestricted model or dataset redistribution.

2.9 Availability of another QMRF for exactly the same model

No other QMRF for erba_classification_v7_er_alpha is known.

3. Defining the endpoint — OECD Principle 1

3.1 Species

Intended biological target: human estrogen receptor alpha (ER α ; ESR1/NR3A1). In the 1,481-compound development set, 1,419 rows explicitly identify *Homo sapiens*; species metadata are absent for 62 PubChem BioAssay rows. Those 62 records were curated as ER α binding records but lack an explicit species field in the model-ready table.

3.2 Endpoint

Binary classification of ER α direct binding: Binding (positive, class 1) versus Non Binding (negative, class 0). This is not ER α functional agonism/antagonism, transcriptional activation, reporter-gene activity, proliferation, or general endocrine-disruption activity.

3.3 Comment on endpoint

The current endpoint is pooled from heterogeneous public direct-binding evidence, including IC₅₀, K_i, RBA, inhibition, binding-affinity and categorical activity records. It is not restricted to one experimental protocol. Consequently, predictions represent consistency with the curated pooled binding class, not an estimate from a single standardized assay. When K-FDA supplies its single-assay dataset, the assay contract should be documented and used to filter compatible evidence; a retrained assay-homogeneous model will require a new model identifier and QMRF.

3.4 Endpoint units

Categorical; no physical unit. Output classes are Binding and Non Binding. The software additionally returns an SVC probability-like score and applies a fixed 0.5 classification threshold. The probability is a model output and must not be interpreted as binding affinity, potency, regulatory certainty, or experimental probability of effect.

3.5 Dependent variable

$y = 1$ for curated positive ER α direct-binding evidence and $y = 0$ for curated negative ER α direct-binding evidence. Development set: 629 positive and 852 negative compounds.

3.6 Experimental protocol

No single experimental protocol describes all records. The model-ready development set includes direct-binding evidence labelled as ER_binding and endpoint labels including IC₅₀ (1,120 rows), K_i (74), categorical Activity (73), RBA (66), PubChem ER α binding dose-response (50), Inhibition (37), log(RBA) (15), EC₅₀ (13), covalent binding (10), PubChem binding affinity (9), and a small number of other binding endpoints. Protocol details remain linked to source records. This heterogeneity is a material limitation.

3.7 Endpoint data quality and variability

Public-source records were curated, structure-standardized, deduplicated to unique parent structures, and resolved to one binary label per retained model-ready structure. The development set comprises 1,221 ChEMBL records, 198 ChEMBL regression-derived positive records, and 62 PubChem BioAssay records. The regression-derived positives were assigned using the ER α positive-match pIC₅₀ interquartile interval [7.50439325, 8.607218]. This derivation and heterogeneous assay provenance can introduce label noise and source bias. Experimental repeatability, inter-laboratory reproducibility, chemical purity, and assay-specific uncertainty are not uniformly available. The current model should therefore be treated as an internal screening model. The future K-FDA single-assay dataset is expected to permit a stricter endpoint definition and prospective validation.

4. Defining the algorithm — OECD Principle 2

4.1 Type of model

Binary fingerprint-based QSAR classification model using a support vector classifier with a radial basis function (RBF) kernel and probability estimation enabled.

4.2 Explicit algorithm

Input SMILES → fixed parent-structure preprocessing → 2,048-bit RDKit topological fingerprint → median imputation → zero-variance feature removal → SelectKBest(k='all') → StandardScaler → scikit-learn SVC(kernel='rbf', C=2.0, gamma='scale', probability=True, random_state=20260811). The fitted effective gamma is 0.0004882812492761384. The fitted classifier uses 710 support vectors (434 class 0; 276 class 1). Class 1 is returned when the positive-class probability is at least 0.5; otherwise class 0 is returned. These fitted-pipeline details are frozen in QMRF_ERalpha_Binding_Classification_v7_supporting_evidence.json.

4.3 Descriptors in the model

2,048 binary hashed RDKit topological fingerprint bits generated with GetRDKitFPGenerator(minPath=1, maxPath=7, useHs=True, branchedPaths=True, useBondOrder=True, countSimulation=False, fpSize=2048, numBitsPerFeature=2) using RDKit 2024.03.5. The fingerprint hashes linear paths and branched topological subgraphs. All 2,048 positions were offered to the final classifier; zero-variance filtering is fitted inside the pipeline. Hashed bits do not have unique human-readable physicochemical names.

4.4 Descriptor selection

Seven feature families were compared during development: RDKit 2D descriptors; Morgan radius 2 and radius 3 fingerprints; RDKit topological fingerprint; MACCS keys; atom-pair fingerprint; topological-torsion fingerprint; and Avalon fingerprint. Each was evaluated with multiple conventional ML algorithms and hyperparameter configurations. Model-family selection was based only on mean outer-fold MCC from development-only nested grouped cross-validation. The selected configuration was rdkit_fp_2048__svm. Within the selected pipeline, SelectKBest retained all post-variance features.

4.5 Algorithm and descriptor generation

SMILES are parsed and sanitized by RDKit. RDKit LargestFragmentChooser() selects one fragment; the retained fragment is rejected unless it contains carbon and only supported elements. The implementation does not search separately for the largest organic fragment. Canonical isomeric SMILES are generated. RDKitFPGenerator creates a deterministic 2,048-bit fingerprint of linear paths and branched topological subgraphs using the settings documented in section 4.3. Preprocessing and feature generation are serialized inside the same pipeline as the classifier to reduce training/inference drift.

4.6 Software name and version

Python 3.10.20; RDKit 2024.03.5; scikit-learn 1.5.2; NumPy 1.26.4; joblib 1.4.2. The broader candidate experiment also had XGBoost 2.1.2 and CatBoost 1.2.10 installed, but the released ER α model is a scikit-learn SVC.

4.7 Chemicals/descriptors ratio

1,481 development chemicals / 2,048 fingerprint positions = 0.723 chemicals per fingerprint position. This low classical ratio is typical for sparse hashed fingerprints but increases overfitting risk; nested grouped cross-validation and regularized SVM training were used to estimate robustness. The final fitted SVC has 710 support vectors.

5. Defining the applicability domain — OECD Principle 3

5.1 Description of applicability domain

The operational AD is the intersection of (a) local distance coverage in standardized RDKit fingerprint space and (b) structural similarity to the ER α development reference. A prediction is labelled In-domain only when both criteria pass. AD is a reliability aid; it was not optimized against an independent external validation set and is not a regulatory validation statement.

5.2 Method used to assess applicability domain

The 1,481 development compounds form the route-specific reference. Reference workbook SHA-256: 761f847d929567679388412bbfed3a19a97fdcde505187dbd8ee4a7369e42620. The ordered row IDs exactly equal development_ERalpha.csv. Their RDKit fingerprints are standardized with StandardScaler. For a query, mean Euclidean distance to the five nearest reference neighbours is calculated. The distance cutoff is the empirical 95th

percentile of each reference compound's mean distance to its five nearest other reference compounds: 54.999514498872124 for the frozen reference. Separately, maximum Tanimoto similarity between the query RDKit fingerprint and reference fingerprints must be at least 0.60. Overall AD = (mean 5-NN distance $\leq$ 54.999514498872124) AND (maximum Tanimoto similarity $\geq$ 0.60). A 2-component PCA projection is provided only for visualization. A Morgan radius-2 nearest-reference structure may be displayed for user interpretation but does not replace the RDKit-fingerprint AD decision. Asset identities and fitted configuration are frozen in the QMRF supporting-evidence JSON.

5.3 Software name and version for AD assessment

Project ADCalculator method knn_mean_distance_p95_rdkit_ad_morgan_display; Python 3.10.20; RDKit 2024.03.5; scikit-learn 1.5.2; NumPy 1.26.4. Route-specific reference: classification_er_alpha_reference.xlsx, 1,481 development compounds; reference SHA-256 761f847d929567679388412bbfed3a19a97fdcde505187dbd8ee4a7369e42620; cache SHA-256 f5b1d657152701e8567c8fdffb41f3774c9ab4d047be4ba1326c426d88a56685. Frozen settings: n_neighbors=5, coverage=0.95, Tanimoto threshold=0.60, fitted distance threshold=54.999514498872124.

5.4 Limits of applicability

- Input must be a valid, sanitizable SMILES structure.
- The largest fragment must contain carbon.
- Allowed atomic numbers in the retained parent are H, B, C, N, O, F, Si, P, S, Cl, Se, Br and I; metals and unsupported elements fail closed.
- Wildcard atoms and CXSMILES syntax are rejected.
- Mixtures, salts and multi-fragment records are represented by the largest selected fragment; activity attributable to a removed component is not represented.
- The model does not establish affinity, potency, agonism/antagonism, in vivo endocrine disruption, or safety.
- The pooled heterogeneous-assay endpoint limits interpretation even for structurally in-domain compounds.

6. Goodness-of-fit and robustness — OECD Principle 4 (internal validation)

6.1 Availability of the training set

The 1,481-row development set is available to the project owner as development_ERalpha.csv and as the route-specific AD reference workbook. Public release/redistribution is subject to source-database terms and scientific-owner approval. It should accompany a regulatory submission under controlled disclosure if permitted.

6.2 Available information for the training set

Available fields include source database, source record locator, compound name where available, CAS where available, PubChem identifiers where available, input and standardized SMILES, InChIKey, receptor subtype, species, target, assay name/class, endpoint, value/unit where available, original and resolved labels, label basis, duplicate group, model SMILES, connected-component group and split fold. Chemical purity and MOL/SDF files are not uniformly available. The dataset contains conventional organic structures, not nanomaterials.

6.3 Descriptor values for the training set

Descriptor values are reproducibly generated from model_smiles by the serialized ERBARawSmilesFeatures transformer. A full 1,481 $\times$ 2,048 fingerprint matrix is not embedded in this narrative document but can be generated and supplied as supporting information together with software/version hashes.

6.4 Dependent-variable values for the training set

Available in development_ERalpha.csv as label/resolved_label and linked label_basis fields: 629 positive and 852 negative.

6.5 Other information about the training set

Development n=1,481 unique parent structures: ChEMBL 1,221; ChEMBL regression-derived 198; PubChem BioAssay 62. All retained structures are classified as organic. Development and sensitivity partitions were split by connected components over exact InChIKey and chiral Bemis–Murcko scaffold, preventing exact-identity or connected-scaffold overlap across partitions.

6.6 Pre-processing before modelling

Blank/invalid structures, wildcards, CXSMILES, no-carbon parents, metals and unsupported elements fail closed. RDKit sanitization is applied, the largest fragment is selected, and canonical isomeric parent SMILES are produced. Duplicate/evidence resolution occurred upstream of modelling, resulting in one retained parent structure per model-ready row. Labels were curated from source categorical binding outcomes and, for 198 development positives, from a documented pIC50-IQR rule. Median imputation, variance filtering, feature selection and scaling are fitted within each training fold through the pipeline.

6.7 Statistics for goodness-of-fit

Resubstitution goodness-of-fit TP, TN, FP, FN, sensitivity and specificity were not computed or frozen and are unavailable; this is an evidence gap. Apparent training-fit statistics were not used for release selection and must not be inferred from section 6.12. Section 6.12 reports grouped cross-validation robustness, not fitted-model goodness-of-fit.

6.8 Leave-one-out cross-validation

Not performed; inappropriate and computationally inefficient for the grouped scaffold/identity design.

6.9 Leave-many-out cross-validation

Five-fold outer / four-fold inner nested StratifiedGroupKFold on the 1,481-compound development set. Groups are connected components over exact InChIKey and chiral Bemis–Murcko scaffold. Within each candidate family, hyperparameters were selected in inner folds by MCC at a fixed 0.5 threshold and outer folds supplied internal scores. Mean outer MCC was then used to select among 56 candidate families (12 parameter configurations per family). Therefore, the selected model's outer metrics are selection-associated internal estimates and are not an unbiased post-selection estimate. Final model hyperparameters were refit on the full development set.

6.10 Y-scrambling

Not performed. This is a documented evidence gap.

6.11 Bootstrap

Not performed. This is a documented evidence gap.

6.12 Other robustness statistics

Selected model outer-fold MCC mean 0.8999, standard deviation 0.0343 (five finite successful folds). These are selection-associated internal estimates because outer-fold mean MCC was used to choose among 56 candidate families; they are not independent post-selection performance estimates. Pooled out-of-fold results: TP 568, TN 839, FP 13, FN 61; sensitivity 0.9030; specificity 0.9847; balanced accuracy 0.9439; F1 0.9388; MCC 0.8987; PR-AUC 0.9554; ROC-AUC 0.9721; Brier score 0.04629. Mean fold PR-AUC was 0.9701. All five folds selected C=2.0, gamma='scale', k='all'. No unresolved warnings or candidate failures were accepted by the release gate.

Outer fold	TP	TN	FP	FN	Sensitivity	Specificity	MCC	PR-AUC
1	116	164	1	8	0.9355	0.9939	0.9372	0.9855
2	161	149	8	8	0.9527	0.9490	0.9017	0.9371
3	38	127	0	5	0.8837	1.0000	0.9221	0.9865
4	112	149	0	25	0.8175	1.0000	0.8367	0.9664
5	141	250	4	15	0.9038	0.9843	0.9018	0.9753

7. Predictivity — OECD Principle 4 (external validation)

7.1 Availability of external validation set

No scientifically independent external validation set is currently available for this exact model. A 224-compound internal resplit exists but is historically exposed to prior project feature generation, training, scoring and model-family analysis; it must not be called external validation.

7.2 Available information for external validation set

Not applicable at present. The internal sensitivity partition has the same field structure as the development set and is retained as internal evidence only.

7.3 Descriptor values for external validation set

Not available because no valid external validation set currently exists.

7.4 Dependent-variable values for external validation set

Not available because no valid external validation set currently exists.

7.5 Other information about external validation set

K-FDA is expected to provide an ER α dataset screened by a single assay method. Unless K-FDA explicitly authorizes training use, that dataset should be sealed before modelling and used once for prospective external validation after the assay contract, endpoints, structures and labels are finalized.

7.6 Experimental design of test set

Planned: K-FDA single-assay dataset, curated independently of the current public-source model development. Exact assay method, receptor construct/species, endpoint, units, active/inactive rule, censoring, replicate handling, exclusions and overlap policy must be registered before opening model predictions. Exact structures/scaffold-connected components overlapping development must be reported and separately analysed.

7.7 Predictivity statistics obtained by external validation

Not available. Required future statistics: TP, TN, FP, FN, sensitivity, specificity, false-positive rate, false-negative rate, balanced accuracy, MCC, F1, PR-AUC, ROC-AUC, Brier score/calibration, coverage, and performance stratified by AD and overlap status. Threshold 0.5 must remain frozen unless a new model/QMRF is created.

7.8 Assessment of external validation set

Pending. Representativeness should be assessed against development chemical space, structural/scaffold overlap, class prevalence, assay protocol, endpoint definition, and AD coverage.

7.9 Comments on external validation

The one-time internal sensitivity evaluation (n=224; 79 positive, 145 negative) yielded TP 62, TN 145, FP 0, FN 17; sensitivity 0.7848; specificity 1.0000; balanced accuracy 0.8924; MCC 0.8381; PR-AUC 0.9511; ROC-AUC 0.9517; Brier score 0.05729. These results are reported only for transparency and are not external/regulatory validation evidence.

8. Mechanistic interpretation — OECD Principle 5**8.1 Mechanistic basis**

The biological endpoint—direct ligand binding to ER α —has a plausible receptor-binding mechanism governed by molecular size/shape, hydrophobic contacts, aromatic topology, hydrogen-bond donor/acceptor patterns and substituent placement. The selected RDKit path fingerprint encodes local/topological substructures relevant to such patterns. However, the nonlinear RBF SVM over hashed bits does not provide a unique causal or readily interpretable mechanistic rule, and fingerprint-bit collisions prevent one-to-one descriptor interpretation. A binding prediction must not be interpreted as agonism, antagonism, transcriptional activity or in vivo endocrine disruption.

8.2 A priori or a posteriori interpretation

Partly a priori at the endpoint level: only curated ER α direct-binding evidence and 2D structure features were targeted. The specific RDKit fingerprint/SVM configuration was selected empirically through nested cross-validation, so descriptor-level interpretation is a posteriori and limited.

8.3 Other mechanistic information

Nearest-reference structures and AD diagnostics are provided to support analogue-based review, not to establish mechanism. Mechanistic interpretation should be supplemented with assay-specific evidence, receptor-binding knowledge and expert review.

9. Miscellaneous information

9.1 Comments

- Intended use: internal screening/prioritization of organic chemicals for potential ER α direct binding.
- Not intended as a standalone regulatory conclusion or replacement for experimental evidence.
- PubChem has two separate roles in the project: a small portion of public training evidence (62 development rows) and optional CAS-to-SMILES lookup in the application. CAS lookup does not change the model or training evidence.
- Database recollection merely for freshness is not planned. Future collection should target assay-compatible measurements demonstrably absent from the existing provenance inventory.
- If K-FDA's single-assay dataset is used to filter/redefine the training endpoint or add same-assay measurements, the resulting retrained classifier is a new model and requires a new QMRF.

9.2 Bibliography

- OECD (2023). (Q)SAR Assessment Framework: Guidance for the regulatory assessment of (Q)SAR models and predictions, Annex 1 QMRF v2.1. <https://doi.org/10.1787/d96118f6-en>
- OECD (2007). Guidance Document on the Validation of (Q)SAR Models. OECD Series on Testing and Assessment No. 69.
- VEGA HUB. QSAR Model Reporting Format (QMRF) of VEGA models. <https://www.vegahub.eu/portfolio-item/vega-qsar-models-qrmf/>
- VEGA. Estrogen Receptor Relative Binding Affinity Model v1.0.2 QMRF. https://www.vegahub.eu/vegahub-dwn/qmrf/QMRF_ESTROGEN_RBA_IRFMN.pdf
- VEGA. Estrogen Receptor-mediated effect (IRFMN/CERAPP) v1.0.1 QMRF. https://www.vegahub.eu/vegahub-dwn/qmrf/QMRF_ESTROGEN_CERAPP.pdf
- RDKit documentation. <https://www.rdkit.org/>
- scikit-learn SVC documentation. <https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>

9.3 Supporting information

Recommended controlled submission package: (1) this QMRF; (2) QMRF_ERalpha_Binding_Classification_v7_supporting_evidence.json, SHA-256 45f28d92104d460be776bbc3790ba5ad4406287b3defe3ae375865d12d407e89; (3) development_ERalpha.csv; (4) nested_cv_results.csv; (5) nested_cv_outer_predictions.csv; (6) internal_resplit_ERalpha.csv and internal_resplit_predictions_ERalpha.csv clearly marked INTERNAL/HISTORICALLY EXPOSED; (7) protocol.json; (8) source_manifest.json; (9) split_manifest.json; (10) production_classification_model_ERalpha_manifest.json; (11) release_manifest_ERalpha.json; (12) preprocessing, feature and AD source files; (13) frozen model and AD asset hash receipts; and, when available, (14) sealed K-FDA prospective external-validation protocol/results. The supporting-evidence JSON binds the exact source/species/endpoint tabulations, pipeline introspection, AD reference/configuration/cutoff and artifact hashes used in this QMRF.

Appendix A — Frozen model and evidence identity

Item	Value
Model ID	erba_classification_v7_er_alpha
Task/subtype	classification / er_alpha
Model file	production_classification_model_ERalpha.joblib
Model SHA-256	db3f2c369cfc9f2a08ecc985424879b70b82f3a4bc8d11a5e2f94ab8c7fe96ea
Model size	11,743,515 bytes
Preprocessing policy	erba_binding_classification_parent_v2
Pipeline schema	raw_smiles_pipeline_schema_v1

Decision threshold	0.5
Evidence scope	internal_historically_exposed
Development partition	1,481 rows; SHA-256 0c042eda221d4055f4e701e8f6c6edddfe6053f57f59514d36c7e7cc957ab486
Internal sensitivity partition	224 rows; SHA-256 4c791e830793ba5db2f504f17033ee4c663d46d97790ca33ba46ba65ff9c8791
Protocol SHA-256	406da317873c99882cab4f07f280ecacca26e1f1fa47d70822e6c446835ffe6d
Release manifest evidence	Evaluator-bound release_manifest_ERalpha.json: release_eligible=true; model SHA-256 and release gates bound to exact evidence.
QMRF supporting evidence	QMRF_ERalpha_Binding_Classification_v7_supporting_evidence.json; SHA-256 45f28d92104d460be776bbc3790ba5ad4406287b3defe3ae375865d12d407e89

Appendix B — QMRF review basis

The document structure follows OECD QMRF v2.1. Content granularity and presentation were compared with the VEGA estrogen-receptor QMRFs listed below. Those QMRFs describe different models and datasets; no performance or method was copied into the present model description.

OECD QMRF v2.1 template: <https://www.oecd.org/chemicalsafety/risk-assessment/qsar-assessment-framework-annex-1-qsar-model-reporting-format.docx>

OECD QAF guidance: https://www.oecd.org/en/publications/q-sar-assessment-framework_9b064821-en.html

VEGA QMRF index: <https://www.vegahub.eu/portfolio-item/vega-qsar-models-qrmf/>

VEGA Estrogen RBA QMRF: https://www.vegahub.eu/vegahub-dwn/qmrf/QMRF_ESTROGEN_RBA_IRFMN.pdf

VEGA ER-mediated effect QMRF: https://www.vegahub.eu/vegahub-dwn/qmrf/QMRF_ESTROGEN_CERAPP.pdf

Appendix C — Scientific-owner approval

Role	Name/signature	Date
QMRF author		
Model developer		
Scientific owner		
Independent reviewer		